Realizable graph moment witness

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1. A graph witness with fixed inverse moments

Let the test input be `u=(e0-e1)/sqrt(2)`. On the 16-vertex rook graph, its positive-Laplacian spectral measure consists of weights `1/4` at `lambda=4` and `3/4` at `lambda=8`, hence transformed inverse-eigenvalue moments

`m0=1, m1=5/32, m2=7/256, m3=11/2048, m4=19/16384`.

Construct a connected four-vertex complete weighted graph with fixed weights `w01=w03=w23=1/50`, and unknown `(w02,w12,w13)`. With `S=[e0-e3,e1-e3,e2-e3]`, `B=(S^T S)^{-1}S^T L S`, `G=S^TS`, `h=(1,-1,0)^T`, and `d=det B`, define three integer/rational polynomial equations

`P_j=0.5 h^T G adj(B)^j h - m_j^(rook) d^j=0`, `j=1,2,3`.

Use the full exact rational centre recorded with the certificate and the radius $r = 10^{-34}$. Let C be the inverse Jacobian at that centre. Exact rational polynomial-coefficient bounds for the Hessians on this box give a contraction factor q and centre displacement ρ satisfying $0 \le q < 1$ and $\rho + qr < r$. Their displayed numerical values are approximately $q = 4.479010085739296 \times 10^{-27}$ and $\rho = 3.689057310400891 \times 10^{-63}$; the proof uses the exact inequalities, not those rounded renderings. Banach's fixed-point theorem therefore gives a unique root in the box. Exact characteristic-polynomial sign changes at the rational centre locate its three positive eigenvalues in $(0.084, 0.086)$, $(5.58, 5.59)$, and $(33.81, 33.82)$. Perturbing the three conductances by at most r changes the Laplacian operator norm by at most $6r$, so the root has spectral gap greater than 0.08. A direct polynomial numerator-and-determinant derivative bound, with a strictly positive determinant lower bound on the box, proves exactly $0.006804564875115 < m_4 < 0.006804564875116$.

The computed root is approximately:

$$\begin{pmatrix} 15.689998458283134606411247 7306 \\ 3.968871903781207382760182 1080 \\ 0.023562856373415166885233 0800 \end{pmatrix}.$$

Its positive eigenvalues are approximately `(0.084748449338,5.585135434187,33.814982553350)`, with pair weights `(3.060828e-7,0.84746316616,0.15253652775)`. The slow mode dominates m4 despite tiny weight: its m4 contribution is approximately `0.005933513`.

Sixteen-site lift. Add 12 new vertices; reduce every old four-site edge weight by `3epsilon` with `epsilon=3/500`, connect every old vertex to every new vertex with weight `epsilon`, and connect all new pairs with weight `1/100`. Every edge stays positive (the smallest old adjusted weight is `0.002`). The cross block annihilates original four-site zero-sum vectors. On those vectors, the old-site block agrees with the original L. Hence the selected pair's entire spectral measure is preserved. The additional eigenvalues are `16epsilon=0.096` (one copy) and `0.12+4epsilon=0.144` (eleven copies), so the original low eigenvalue remains the 16-graph spectral gap. Both candidate graphs have 16 vertices, matching m0..m3 for the one specified pair but differing at m4.

This is **exact conditional nonidentifiability** from four transformed moments of one pair, not complete-data indistinguishability, a new physical law, or a first-in-literature claim.

2. A threshold measurement for the graph witness

Let `g_lambda(s)=4lambda^2 s exp(-2lambda s)` be the calibrated scale density for a positive eigenmode. Rook-edge density:

p_R(s)=(1/4)g_4(s)+(3/4)g_8(s).

Alternative density:

p_A(s)=sum_{j=1}^3 w_j g_(lambda_j)(s) with the certified eigenvalue and weight intervals.

Numerical evaluation of the full distributions gives numerical total-variation distance 0.104957402798 (four detected density crossings) and Bhattacharyya coefficient 0.990246324123. These latter integrals are high-accuracy numerical calculations, not outward-certified integral bounds.

A separate **fully certified** binary experiment requires much less: record the bit Y=1{s>3/100}. The exact rational graph eigenvalue/weight enclosures together with rational Taylor bounds for exponentials prove

P_R(Y=1)>930704/1000000, P_A(Y=1)<870032/1000000.

For p>q, the binary Bhattacharyya coefficient sqrt(pq)+sqrt((1-p)(1-q)) is maximized by reducing p and increasing q. Exact rational square comparisons then show B<994759/1000000. Tensorization and the classical Bhattacharyya bound give

P_e(M) <= (1/2) B^M < (1/2)(994759/1000000)^307 < 1/10.

Thus **307 independent photon trials suffice for <=10% equal-prior Bayes error** under this exact scale-threshold measurement contract, for this graph pair, with no detector loss and calibrated known source. This is a conservative sufficient guarantee, not the minimum and not a general physical resource bound.

The submitted 57,291,466 Cantelli guarantee concerns **a different** three-atom positive-measure witness *and* the fourth-moment estimator. Comparing those trial counts as if they were the same alternative/model/estimator would be invalid.

This illustrates a central lesson: matching several moments does not imply closeness of complete distributions. The tiny slow-mode spectral weight makes m4 sensitive to rare long-scale outcomes, while much larger spectral-weight rearrangements cause readily testable differences at short scales.

3. Statistical threshold calculations

The simple equal-phase and antisymmetric rook-edge/nonedge scale densities give the following tests:

- For a single identically repeated upper-threshold bit, equal-phase pair: M=165, t=0.22943, risk 0.099786766829, whereas the numerically optimized M=164 risk is ~0.10046967.
- Antisymmetric pair: M=173, t=0.23077, risk 0.099493720411; numeric optimized M=172 risk ~0.10017110.
- The source's **exact rational sweep certificates** exclude every upper threshold at M=164 and M=172, respectively. The threshold distributions were checked numerically. The claimed global minimum rests on the supplied exact interval-sweep certificate; numerical agreement alone is insufficient.
- Fourier-likelihood inversion reproduces M=116:0.10022951, M=117:0.09926536 for equal phase, and M=126:0.10028047, M=127:0.09939238 for antisymmetric. These full-distribution crossings are **numerical**.
- For the antisymmetric sample-mean Chernoff statistic at threshold 0.1711, the prescribed positive and negative exponential tilts give 0.0998551584 and 0.0998533841 at M=375. The source supplies stronger rational exponential enclosures; The reported floating values agree with those enclosures.

4. Quantum information, phase access and calibration rank

For a_lambda(s)=2lambda sqrt(s) e^(-lambda s) in L^2(ds), the pure-state quantum Fisher information for lambda equals 2/lambda^2; classical measurement of s achieves the same FI. Multiplying the amplitude by a *parameter-dependent* phase exp(i b lambda s) preserves all scale probabilities but changes the QFI to 2(1+b^2)/lambda^2. Thus coefficient two is **not** a universal quantum-physical ceiling. This agrees with Braunstein–Caves 1994. The phase dependence is an additional physical instrument assumption, not a free resource.

Maximizing the specified three-port real-amplitude local-FI objective along equal scales gives `0.8279358531658`. For the quoted equal-scale alias parameters, the ideal continuous-register overlap is `0.3429323662853`, one ideal pure photon has optimal error `0.0303199514`, and a coherent known-amplitude input requires mean energy `0.7774323335` to reach 10% error. These apply to distinct measurement contracts. The source report contains an analytic global-optimum proof of the three-port class; the numerical calculation checks its value and does not replace the analytic global-uniqueness proof.

On the rook and Shrikhande graphs, take `k=128L^+=24I+4A-3J`. The exact calculation constructs all 120 contrast-perturbation residual derivatives, collects 120 off-diagonal components per derivative, and eliminates the resulting **integer** `120 x 120` matrix modulo `1,000,003`. Both ranks equal **119**. Both matrices have zero column sums. This proves rank >=119 over rational numbers and rank <=119 from the exact column sum identity, hence **rational rank=119**. It supports the revised distinction between `d-1` constraints for common spectral functions and locally 119 directions for pair-specific contrast perturbations.

For the deterministic power-law readout $S' = \kappa S^p$, the transformed inverse-moment relation persists under the stated common coherence contrasts; consequently a zero quadratic residual alone does not identify linear scale calibration. A classification of all measurable readout functions preserving that graph-moment relation requires a separate argument. The controlled density-ratio theorem in *Detector Transfer Ambiguity* instead addresses recovery of the dimensionless incident coordinate under one common standard-Borel subprobability response kernel, known control multipliers and per-preparation normalization. Its zero-conditional-variance criterion does not establish the power-law classification.